

LUIS DANIEL DOS SANTOS

AI & SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of production experience, specializing in RAG pipelines, agentic systems, and model optimization for commercial clients. Combines a UBA law degree with software engineering to apply formal logic and deterministic verification to probabilistic AI models. Native Spanish, Full Professional English, Elementary Japanese (JLPT N5, daily study).

CORE SKILLS

- **Agent Engineering:** LangGraph - MCP Servers & Tool Calling - State Graphs & Circuit Breakers - Agent Evals (Langfuse, Promptfoo)
- **AI & LLM:** RAG Pipelines - Fine-Tuning & Model Distillation - SLMs & Quantization - Vector DBs (pgvector, Qdrant)
- **Backend & Systems:** Python - TypeScript - Rust - FastAPI - NestJS - WebSockets & REST APIs - PostgreSQL - Docker - Kubernetes - CI/CD
- **Testing & Architecture:** Automated Testing (Pytest, Jest) - GitHub Actions CI/CD - System Architecture - Edge-Case Modeling - Auditability-by-Design

WORK EXPERIENCE

AI SYSTEMS ENGINEER

Independent Practice — Remote / Tokyo

03/2026 – Present

- Engineered domain-specific RAG pipelines and structured data extraction engines for 6+ international clients across legal and commercial operations.
- Developed headless automation services in Python and Rust using Playwright; automated repetitive browser workflows, saving 15+ hours weekly per client.
- Managed technical delivery from system architecture through containerized deployment; scaled the practice to 6+ concurrent clients via direct referrals.

LEAD AUTOMATION ENGINEER

Independent Consulting — Buenos Aires

10/2024 – 11/2025

- Engineered RPA pipelines and web scrapers in Python and Playwright, scaling automated legal workflows to 30+ concurrent commercial clients.
- Cut administrative submission turnaround by 70% with a 0% rejection rate across 300+ submissions to public registries and state platforms.
- Deployed document indexing pipelines using Mistral OCR to digitize physical case files; cut intake pre-classification time by 40%.
- Authored commercial contracts, NDAs, and compliance reviews, integrating deterministic validation rules directly into regulatory & compliance workflows.

AI & SOFTWARE ENGINEER (CONTRACT & RESEARCH)

Multi-Client Portfolio (Concurrent with Law Degree at UBA)

11/2022 – 03/2025

- **FinTech Algorithmic Engine (Bots4Hodl)**: Developed an RNN-LSTM pipeline for real-time market forecasting over WebSockets and integrated a multi-currency payment gateway using PyTorch, FastAPI, and Docker.
 - **Speech AI Optimization (UBA Law Lab)**: Fine-tuned OpenAI Whisper on legal trial records and modified the decoding pipeline for CPU-only inference; achieved 96% domain accuracy without GPU dependencies.
 - **Computer Vision System (Retail Chain)**: Deployed a shelf-stock and price verification service with YOLO, OpenCV, and FastAPI; cut per-item inventory audit times from several minutes down to ~1 second.
 - **Conversational AI Assistant (Hospitality Client)**: Architected a 24/7 guest booking agent using WhatsApp Business API and RAG workflows; automated two-way calendar sync, eliminating booking overlaps across customer channels.
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RELEVANT PROJECTS

CORTEX-GATE (Rust, Tokio, Axum, Tauri v2) — github.com/luisdossantos-dev/cortex-gate

- LLM proxy routing requests between SLMs and cloud models via rule-based lexical routing; runs on CPU in <1.2ms (<25MB RAM) and reduced API infrastructure costs by 60%.

KELSENGRAPH (Python, NetworkX, Pydantic, DAGs) — github.com/luisdossantos-dev/kelsen-graph-poc

- Formal verification engine mapping statutory rules to DAGs; resolves logical rule conflicts deterministically in <15ms, enforcing mathematical consistency over probabilistic model outputs.

PAYSUITE (TypeScript, NestJS, pgvector, LangGraph.js) — github.com/luisdossantos-dev/PaySuite

- Transaction scoring backend with a two-tier fraud engine (sub-25ms memory rules, <180ms P95 LLM evals) and sub-millisecond HNSW vector search over 1M+ records.

DISTILKIT (Python, PyTorch, ONNX Runtime, SLMs) — github.com/luisdossantos-dev/distilkit

- Knowledge distillation and ONNX graph optimization pipeline for SLMs; cut model size and VRAM by 68% while increasing inference throughput by 4.1x on CPU runtimes.

LEGIS-AGENT (FastAPI, LangGraph, Tauri v2, spaCy, Promptfoo) — github.com/luisdossantos-dev/legis-agent

- Desktop audit client using local spaCy NER to redact PII (APPI/GDPR compliant) before model ingestion, monitored with Langfuse and Promptfoo CI evals.
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EDUCATION & ACADEMIC BACKGROUND

BACHELOR OF LAWS (LL.B.)

Universidad de Buenos Aires (UBA)

Graduated 2025

- Applied formal statutory logic to deterministic software design, mitigating runtime edge cases.

COMPUTER SYSTEMS & LOGIC INSTRUCTOR

Academic Tutoring & Student Support

03/2020 – 12/2025

- Tutored university students in formal logic, memory constraints, and low-level computer systems fundamentals.

CONTINUING EDUCATION

Independent Technical Research

01/2017 – Present

- Progressed from self-taught computing fundamentals and Linux systems into applied research across distributed architectures, model optimization, and neuro-symbolic AI.